

Dataset Preparation for AlexNet using Tiny-ImageNet data

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Submitted by Chandra Parasa

1. Getting Tiny ImageNet Dataset

The Tiny ImageNet dataset is a widely used benchmark dataset in computer vision research, designed to be a smaller and more manageable subset of the full ImageNet dataset. It contains 200 different classes, each represented by 500 training images, 50 validation images, and 50 test images. All images are colored RGB images with a resolution of 64×64 pixels.

For this assignment, the dataset was downloaded as a compressed ZIP file and extracted into a designated working directory. After extraction, the directory structure was carefully verified to ensure that:

- All 200 class folders were present in the training directory.
- Each class folder contained the expected number of images.
- The validation and test folders were correctly structured.

Verifying the dataset integrity at this stage is critical to avoid errors during subsequent preprocessing and splitting. Ensuring the full dataset is available provides a solid foundation for creating a reproducible, class-balanced subset.

2. Creating own Dataset from the Tiny ImageNet Dataset

To train convolutional neural networks effectively, it is important to work with a dataset that is manageable in size, class-balanced, and properly split. Therefore, a custom subset of Tiny ImageNet was created following the steps below.

2.1. Considering Only 100 Classes from the Dataset

From the original 200 classes, 100 classes were randomly selected to create a representative subset. Random selection was used to avoid introducing bias in class selection and to simulate a typical real-world scenario where all classes are not always used. A random seed was set to ensure reproducibility, so the same subset of 100 classes would be selected in any repeated experiment.

Sorting the selected classes after sampling ensures a consistent order, which is useful when mapping class names to numeric labels for model training and evaluation.

2.2. From Each Class, taking 500 Images

For each of the 100 selected classes, 500 images were chosen exclusively from the original training dataset. Using only the training images ensures full control over the dataset splits and avoids relying on the original validation or test sets, which may have uneven class distributions or small numbers of images.

Before selection, the images were randomly shuffled to:

- Introduce diversity, so that the subset is representative of the full class.
- Avoid any potential ordering bias that could occur if the original images were arranged in a particular sequence.

Only the first 500 images after shuffling were retained, guaranteeing that each class contributes equally to the custom dataset.

2.3. Splitting Dataset into Training, Validation, and Test Sets

The 500 images from each class were divided into three non-overlapping subsets:

- **Training set:** 300 images per class
- **Validation set:** 100 images per class
- **Test set:** 100 images per class

This results in the following total number of images:

- Training set: $100 \times 300 = 30,000$ images
- Validation set: $100 \times 100 = 10,000$ images
- Test set: $100 \times 100 = 10,000$ images

Details of the splitting process:

1. All 500 images per class come exclusively from the original training directory.
2. Images are first shuffled randomly to ensure variability and diversity in each split.
3. The first 300 images are assigned to the training set.
4. The next 100 images are assigned to the validation set.
5. The final 100 images are assigned to the test set.
6. This guarantees that the three subsets are completely disjoint, with no image appearing in more than one set.
7. Each class contributes equally to all splits, ensuring class balance.

By using only the training images and performing a controlled split, we maintain full reproducibility and consistency, which is critical for proper model training and evaluation.

2.4. Preparing Data for the Model as described in Section 2 of the AlexNet Paper

The dataset was prepared following the preprocessing and input requirements outlined in Section 2 of the AlexNet paper [1], ensuring consistency with AlexNet’s data handling principles while adapting them for the Tiny ImageNet dataset.

- **Training set:** To enhance model generalization, images were augmented through random cropping and horizontal flipping. Each image was then converted to a tensor and normalized using the ImageNet mean and standard deviation values. These steps help the model learn robust features and reduce the risk of overfitting.
- **Validation and Test sets:** Images in the validation and test sets were converted to tensors and normalized using the same mean and standard deviation, but without augmentation. This ensures consistent evaluation conditions and provides a reliable measure of model performance.
- **Data storage:** The processed images were saved both in a class-wise folder structure for visual verification and as NumPy arrays for efficient loading during model training.

Unlike the original AlexNet implementation, which uses input images of size 224×224 , the Tiny ImageNet dataset provides images at a native resolution of 64×64 . The images were intentionally not resized to 224×224 to preserve their original structure and reduce unnecessary computational overhead. Resizing to a larger resolution would increase memory consumption and training time without offering meaningful benefits, given the limited detail available in Tiny ImageNet images. Instead, the model architecture was adjusted to handle 64×64 inputs directly, maintaining efficiency while remaining conceptually aligned with AlexNet’s design. This preprocessing pipeline ensures that the dataset is optimized for use with the adapted AlexNet model, maintaining both computational efficiency and consistency with the principles described in the original paper.

Sample images from various classes across the training, validation, and test sets are visualized after performing the data split, as shown in Figure 1.

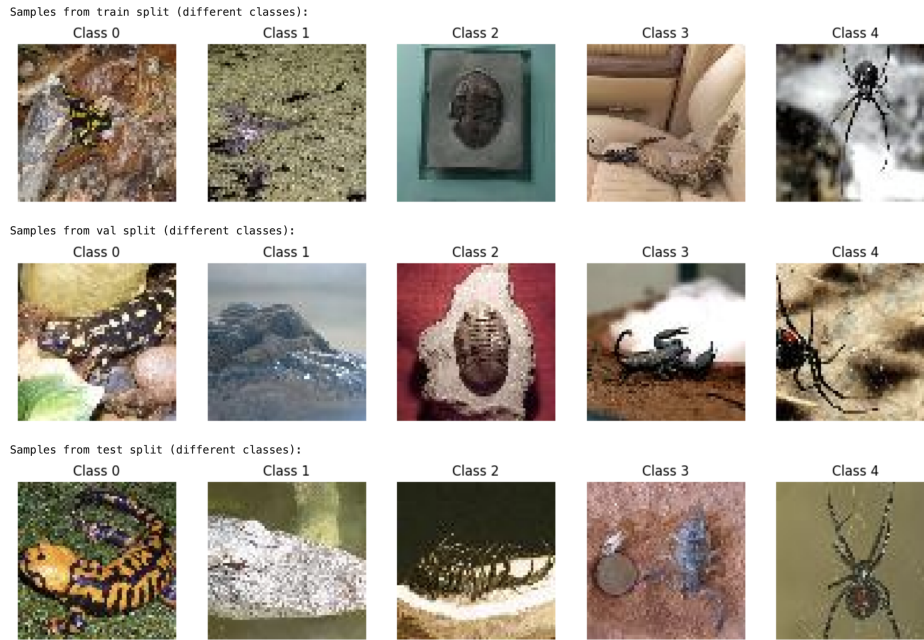


Figure 1: Sample images representing various classes from the training, validation, and test datasets after splitting.

3. Summary

In summary, the custom Tiny ImageNet dataset created for this assignment meets all the specified requirements:

- Contains 100 randomly selected classes from the original 200 classes.
- Each class contributes 500 images taken exclusively from the training dataset.
- Training, validation, and test sets contain 30,000, 10,000, and 10,000 images respectively.
- All splits are class-balanced and non-overlapping.
- Preprocessing is performed according to AlexNet specifications, including augmentation for training and normalization for all splits.
- The dataset is stored in both class-wise folders and NumPy arrays for efficient access and reproducibility.

This carefully prepared dataset provides a solid foundation for training and evaluating convolutional neural networks, ensuring reproducibility, consistency, and appropriate data representation.

References

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, pp. 1097–1105, 2012.